Expected Outcomes for Energy-Flow Cosmology Cross-Survey Invariance and BIG-SPARC Tests

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Internal working note. Pre-data estimation based on structural analysis.

DOI: 10.6084/m9.figshare.32045592

Abstract. This note estimates expected outcomes for the pre-registered EFC cross-survey invariance test (8 sealed predictions, P1–P8) prior to data analysis. The estimation uses the known SPARC-175 Kill-Test v6 baseline (60.2% win rate, $k = 0.415 \pm 0.029$), structural factors affecting cross-survey transfer, and the BIC/AIC distinction to project realistic win rates, overlap consistency, and regime fractions for THINGS (34 galaxies), LITTLE THINGS (26 galaxies), and WALLABY DR2 (203 galaxies). The most likely outcome is Scenario B (partial support, 50–55% probability), where Tier-1 non-SPARC data are suggestive but underpowered ($N = 60$), and BIG-SPARC becomes the decisive tiebreaker.

1 Known Baseline

The SPARC-175 Kill-Test v6 universality analysis (DOI:10.6084/m9.figshare.31986762) provides the calibration anchor:

Table 1: SPARC-175 Kill-Test v6 baseline

Metric	Value
Galaxies fitted	171 / 175
EFC win rate (AIC)	60.2%
Decisive ($\lvert\Delta\text{AIC}\rvert > 10$)	72
Marginal ($0 < \lvert\Delta\text{AIC}\rvert \leq 10$)	31
Tied	46
ΛCDM wins	22 (6 marginal + 16 decisive)
Median ΔAIC	+6.2
Median χ^2_{red} (EFC)	0.44
Median χ^2_{red} (NFW)	1.69
Regime proxy fractions (chi2-ratio method):	
FLOW	71 / 171 = 41.5%
TRANSITION	84 / 171 = 49.1%
LATENT	16 / 171 = 9.4%
Screening parameter	$k = 0.415 \pm 0.029$

Key structural feature: the 72 decisive EFC wins have $\Delta\text{AIC} < -10$. These are robust against pipeline differences and BIC penalties. The 31 marginal wins ($\Delta\text{AIC} \in [-10, 0)$) are the vulnerable population.

2 Structural Factors Affecting Transfer

Six factors degrade cross-survey performance relative to the SPARC baseline, and two factors partially compensate.

Table 2: Structural factors and estimated win-rate shifts

Factor	Tier	Win-rate shift
No Spitzer mass decomposition (THINGS/LT)	T1	-5%
Different velocity extraction pipeline	T1	-3%
HI-only (no Hα inner points)	T1	-2%
WALLABY beam smearing	T2	-8%
WALLABY assumed 10% fractional errors	T2	-4%
Smaller sample $\rightarrow$ more variance	T1	$\pm0\%$
THINGS high angular resolution	T1	$+2\%$
LITTLE THINGS dwarfs (FLOW-dominated)	T1	$+3\%$
Net shift (Tier-1)		$-$**5**%
Net shift (WALLABY)		$-$**12**%

Critical caveat: The -5% Tier-1 shift is dominated by the missing mass decomposition. Without Spitzer photometry, the Kill-Test compares fit quality (rotation curve shape) rather than the full EFC screening physics. This is an indirect test of k, not a direct measurement.

3 Estimated Win Rates

3.1 AIC-Based Estimates

Table 3: Estimated AIC win rates with 95% Wilson confidence intervals

Survey	N	WR (AIC)	95% CI	Pre-reg threshold
SPARC (known)	171	60.2%	[52%, 67%]	—
THINGS	34	$\sim$57%	[41%, 72%]	—
LITTLE THINGS	26	$\sim$60%	[41%, 76%]	—
WALLABY	203	$\sim$43%	[37%, 50%]	$\geq$ 45% (P4)
Tier-1 non-SPARC	**60**	$\sim$**58%**	**[46%, 70%]**	$>$ 55% strong

Key observation: The Tier-1 CI of [46%, 70%] is too wide to distinguish 58% from 50% at 95% confidence. With $N = 60$, any win rate between 45% and 70% is statistically plausible even if the true rate is 58%. This is why Scenario B (partial support) is the most likely outcome.

3.2 BIC Impact

BIC penalises additional parameters more heavily than AIC: $\Delta\text{BIC} = \Delta\chi^2 + \ln(N) \times \Delta k_{\text{params}}$ versus $\Delta\text{AIC} = \Delta\chi^2 + 2 \times \Delta k_{\text{params}}$. The extra penalty per galaxy is $\ln(N_{\text{pts}}) - 2$:
Interpretation: If BIC Tier-1 win rate holds above 50%, the EFC advantage is not driven by extra parameter flexibility — it reflects a genuine fit-quality improvement. If BIC flips the

Table 4: BIC penalty and estimated win-rate reduction

Survey	Typical N_{pts}	Extra penalty	WR (AIC)	WR (BIC)
SPARC	20	+1.0	60.2%	∼58%
THINGS	25	+1.2	∼57%	∼51%
LITTLE THINGS	15	+0.7	∼60%	∼56%
WALLABY	12	+0.5	∼43%	∼35%
Tier-1 non-SPARC	—	—	**∼58%**	**∼53%**

result below 50%, the signal sits in the 31 marginal SPARC galaxies where the extra EFC parameter is absorbing noise, not structure.

4 Prediction-by-Prediction Estimates

Table 5: Expected outcomes for all 8 pre-registered predictions

ID	Prediction	Estimate	Threshold	Expected verdict
P1	k invariance (win rate proxy)	$\sim$58%	$> 55\%$	MARGINAL PASS
P2	Overlap consistency (14 gal.)	10–12/14	$\geq 10/14$	LIKELY PASS
P3	Regime fractions (± 15 pp)	Within range	± 15 pp	LIKELY PASS
P4	WALLABY win rate	$\sim$43%	$\geq 45\%$	BORDERLINE
P5	$k_{\mathrm{BIG}} = 0.415 \pm 0.015$	Awaiting data	3σ	PENDING
P6	BIG-SPARC global WR $\geq 50\%$	Awaiting data	50%	PENDING
P7	FLOW-regime WR $\geq 85\%$	Awaiting data	85%	PENDING
P8	No secondary k dependence	Awaiting data	3σ	PENDING

P1 and P4 are the two predictions closest to their thresholds. P2 and P3 are structurally robust because decisive wins ($|\Delta\mathrm{AIC}| > 10$) are insensitive to pipeline differences. P5–P8 cannot be estimated because BIG-SPARC data are not yet released.

5 Overlap Test (P2) — Detailed Assessment

Fourteen galaxies appear in both SPARC and THINGS: NGC 2403, NGC 2841, NGC 2903, NGC 2976, NGC 3198, NGC 3521, NGC 3621, NGC 4736, NGC 5055, NGC 6946, NGC 7331, NGC 7793, DDO 154, and IC 2574.

These span the full regime range: DDO 154 and IC 2574 are FLOW-regime dwarfs (strong EFC wins on SPARC); NGC 2841 and NGC 5055 are LATENT-regime massive spirals (LCDM wins on SPARC). For decisive cases, the verdict is pipeline-insensitive and $|\Delta\mathrm{AIC}_{\mathrm{SPARC}} - \Delta\mathrm{AIC}_{\mathrm{THINGS}}| < 3$ is expected. The 2–4 marginal cases (TRANSITION regime) are where divergence is most likely due to: (1) different inner-radius coverage (SPARC has Hα; THINGS is HI-only); (2) different beam sizes; and (3) slightly different inclination corrections.

Estimate: 10–12 of 14 consistent. The risk is losing 1–2 marginal cases, which would put P2 exactly at the threshold.

6 Scenario Probabilities

Based on the structural analysis above:

Table 6: Scenario probabilities and characteristics

Scenario	Probability	Signature
A: Universality	25–30%	Tier-1 WR $>$ 55% (CI lower $>$ 40%); $\geq$ 12/14 overlap; k-surrogate stable across Tier-1; KS $p >$ 0.10. *Action: publish, proceed to BIG-SPARC.*
B: Partial support	50–55%	Tier-1 WR 50–58% (CI spans 50%); WALLABY degrades; 10–11/14 overlap; BIC reduces margins. *Action: report Tier-1/Tier-2 separately; BIG-SPARC is tiebreaker.*
C: Falsification	15–20%	Tier-1 WR $<$ 35% (CI upper $<$ 50%); $<$ 7/14 overlap; k-surrogate diverges $> 3\sigma$; KS $p < 0.01$. *Action: document in ledger as FAIL. Revise galactic sector.*

Why Scenario B dominates: With $N = 60$ Tier-1 galaxies, the statistical power to distinguish a true 58% win rate from 50% is low (power ≈ 0.30 for a one-sided binomial test at $\alpha = 0.05$). The test is designed to catch strong signals (Scenario A or C). Scenario B is the "inconclusive but informative" zone that channels the decision to BIG-SPARC.

7 What BIC Reveals That AIC Does Not

If the cross-survey test is run with both AIC and BIC, four diagnostic outcomes are possible:

Table 7: AIC/BIC diagnostic matrix

AIC result	BIC result	Interpretation
WR $>$ 50%	WR $>$ 50%	Strong. EFC advantage is not from parameter flexibility.
WR $>$ 50%	WR $<$ 50%	Marginal. Signal sits in marginal wins where extra parameter matters.
WR $<$ 50%	WR $<$ 50%	Fail. Model loses on both metrics.
WR $<$ 50%	WR $>$ 50%	Impossible (BIC is strictly more conservative than AIC).

Most likely outcome for Tier-1: AIC ∼58%, BIC ∼53% — both above 50%, suggesting the signal is real but modest.

Most likely outcome for WALLABY: AIC ∼43%, BIC ∼35% — AIC already below threshold, BIC confirms. This is expected from Tier-2 data quality, not a physics failure.

8 Decision Framework After Data

1. **First:** Read Tier-1 non-SPARC win rate + Wilson CI. If CI lower bound $> 50\% \rightarrow$ strong support. If CI spans 50% $\rightarrow$ underpowered (Scenario B). If CI upper bound $< 50\%$ $\rightarrow$ falsification (Scenario C). *The primary decision is based on whether the 95% Wilson confidence interval excludes 50%, not the point estimate alone. A point estimate of 58% with CI [42%, 72%] is not evidence of support — it is indistinguishable from chance.*
2. **Second:** Compare ΔAIC and $\Delta\chi^2$ signs. If $> 20\%$ of galaxies show sign disagreement $\rightarrow$ parameter penalty is masking a physics loss. ΔAIC *thresholds follow the standard Burnham & Anderson (2002) interpretation:* $|\Delta\text{AIC}| < 2 =$ *weak evidence;* 2–6 $=$ *moderate;* $> 10 =$ *strong. Values in the 0–2 range should not be interpreted as meaningful model preference.*
3. **Third:** Overlap test (P2). $\geq 10/14 \rightarrow$ physics signal. $< 7/14 \rightarrow$ pipeline artefact (strongest falsification).
4. **Fourth:** k-surrogate stability across Tier-1 surveys. Remember: this is a proxy, not physical k. Stable $\rightarrow$ consistent model shape. Divergent $\rightarrow$ investigate.
5. **Last:** WALLABY. Report separately. Do not combine with Tier-1 for primary conclusions.

9 Limitations of This Estimation

1. All win-rate shifts are heuristic, not derived from simulation. Actual pipeline differences may be larger or smaller.
2. The k-surrogate ($r_{\text{turn}}/r_{\text{max}}$) is not necessarily monotonic with physical k. It is used for cross-survey consistency only, not absolute calibration.
3. Regime proxy fractions use χ^2-ratio, not the original Kill-Test v6 latent variable L. Direct comparison with SPARC-175 regime fractions is approximate.
4. BIC estimates assume a fixed marginal-flip fraction of $\sim$35%. The actual fraction depends on the ΔAIC distribution tail, which varies per survey.
5. No MCMC posteriors exist for any EFC fit. All comparisons are point estimates. Full Bayesian model comparison (nested sampling) remains the most important open gap in the EFC programme.

References

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